Technology & AI Architecture Overview

How Specialised AI, Agentic Reasoning & Network Physics Work Together

Carrier operations cannot rely on static rules alone—which break on noisy optical waveforms, 10,000-alarm multi-vendor storms, and unstructured RFPs/permits—nor on ungrounded chatbots that hallucinate network coordinates. We combine **six purpose-built AI capabilities** with a **deterministic spatial and optical physics core**.

End-to-End Intelligence Loop

1. SENSE (SIGNAL ML)
Waveform & PM Models
Detects digging vibration, Pre-FEC BER cliffs & .sor break vs. bend signatures on live streams.

2. CORRELATE (GRAPH ML)
Causal & Bayesian RCA
Collapses multi-OEM alarm storms across 6 layers and scores root-cause probabilities.

3. INVESTIGATE (AGENT + RAG)
Tool-Use & Vector Memory
Runs diagnostic tool calls (OTDR, PostGIS, CRQs), retrieves SOPs & parses RFP/permit docs.

4. ACT (GATED SYNTHESIS)
Cited Briefs & Approvals
Drafts cited EN/AR RFOs and stages optical reroutes & field dispatches for NOC sign-off.

The Six AI & Machine Learning Layers

1. Agentic NOC Investigation Orchestrator
Tool-Calling ReAct / LangGraph · Claude Sonnet 4, GPT-4o or Llama 3.3 70B (vLLM)

Acts as an autonomous L2/L3 triage agent: dynamically decides which diagnostic tools to invoke—triggering on-demand VIAVI OTDR shots, querying PostGIS slack offsets, checking ServiceNow change windows, and staging disjoint optical reroutes—pausing at human approval gates before execution.

2. Optical, Acoustic & Power Time-Series ML
PyTorch / scikit-learn / ONNX Runtime · 1D-CNN, Isolation Forest & Slope Forecasters

Classifies Phase-OTDR soil acoustics (`18–65 Hz` excavator impacts vs. street traffic) paired with coherent `SOP (rad/s)` spikes for pre-cut warnings; classifies `.sor` trace anomalies (clean break vs. macro-bend pinch); predicts Pre-FEC BER time-to-cliff and 5G battery time-to-dark.

3. Probabilistic Cross-Layer Root-Cause Scoring
Topology-Guided Graph Clustering + Bayesian / Gradient-Boosted Classifiers

Partitions asynchronous 10,000-alarm bursts across OSP ducts, DWDM, IP/MPLS, and 5G RAN, then combines topology evidence with historical alarm co-occurrence to output calibrated probabilities (e.g., 97% trunk cut vs. GPON dying-gasp power blackout or E-band rain fade).

4. Hybrid Telecom RAG & Incident Case Memory
PostgreSQL pgvector + pg_trgm · BAAI/bge-m3 or text-embedding-3 + Reranker

Combines dense semantic embeddings with exact telecom alarm-code lexical matching and cross-encoder reranking to retrieve approved carrier SOPs, vendor manuals (Ciena, Nokia, Ericsson), and similar past resolved tickets (`MTTR` benchmarks) in `<50 ms`.

5. Document & Multimodal Intelligence (RFPs & Permits)
Vision & Document LLMs · Claude Sonnet 4 Vision / Qwen2.5-VL (On-Prem)

Extracts customer A/Z site pairs, diversity mandates, and latency SLAs from unstructured enterprise RFP documents to feed automated Zero-SRLG route design; parses municipal civil roadwork PDFs/notices into geofenced risk zones; validates field closure evidence.

6. Schema-Constrained Multilingual Generation
Pydantic Structured Outputs · Citation-Locked EN / Arabic Synthesis

Generates operator handover briefs, TIA-598-D colour-coded splice packs, and SLA-shielded customer RFOs (automatically separating carrier MTTR from force-majeure or customer-power downtime) in English and Arabic—every sentence bound to verified `[Alarm/Manhole/Runbook ID]` citations.

Why Operators Trust It: Clear Boundary Between AI & Network Physics

OPERATIONAL TASK	EXECUTION ENGINE	HOW PRECISION & SAFETY ARE GUARANTEED
Street Dig GPS & Slack Math	Deterministic Physics	Binary `SR-4731 (.sor)` parser + `PostGIS` trench interpolation subtract manhole slack coils & cable twist. **AI never estimates coordinates.**
Diverse Optical Route Design	Deterministic Graph	`NetworkX` computes primary/backup paths with zero shared ducts, chambers, or bridges. **AI never invents network paths.**
Pre-Cut & Fault Classification	Signal & Graph ML	Time-series & probabilistic models detect dig vibration, BER cliffs, and false-dispatch signatures (power/weather) noisy thresholds miss.
Triage, SOP Lookup & RFP Parsing	Agent + Hybrid RAG	Tool-calling agent chains diagnostics, retrieves `pgvector` SOPs, and extracts structured site/permit data from PDFs & spreadsheets.
Customer RFOs & Network Actions	Cited LLM + Human Gate	LLMs draft cited EN/AR reports from verified JSON; optical switches & ServiceNow dispatches execute **only after NOC engineer approval**.

Platform Infrastructure Stack

LAYER	TECHNOLOGIES	PRODUCTION ROLE
AI & ML Runtime	`Anthropic / OpenAI SDK`, `vLLM`, `ONNX`, `pgvector`	Cloud frontier APIs or air-gapped on-prem open-weight LLMs (`Llama 3.3 / Qwen 2.5`), local embeddings (`bge-m3`), and sub-10ms signal inference.
Backend & Codecs	`Python 3.11+`, `FastAPI`, `Pydantic v2`, `NetworkX`	Async agent/tool orchestration, 6-layer causal graph engine, and native binary `Telcordia SR-4731 (.sor)` + CRC-16 parser.
Spatial & Relational DB	`PostgreSQL 16`, `PostGIS 3.4`, `pg_trgm`	Normalized 6-layer inventory (ducts to SLAs), `WGS84` geospatial indexing, hybrid vector/lexical runbooks, and immutable audit log.
Streaming & Listeners	`Redpanda` (Kafka API), `asyncio` UDP	High-burst telemetry & alarm ingestion (`:19092`) plus native `RFC 5424` Syslog (`:15514`) and `SNMPv2c` traps (`:15162`).
Web & GIS Console	`React 18`, `TypeScript 5`, `MapLibre WebGL`	60 fps street duct/chamber GIS map, interactive OTDR waveforms, 5-layer blast-radius graph, and 1-click NOC approval workspace.

Southbound & Northbound Telecom Interfaces

DOMAIN	SUPPORTED VENDOR SYSTEMS	STANDARD WIRE PROTOCOLS
Optical & OTDR	Ciena MCP, Nokia NSP, VIAVI ONMSi, EXFO Nova	`RESTCONF`, `OpenConfig gNMI`, `TMF642`, Binary `.sor`
OSP GIS & Maritime	Lepton SmartInventory, ESRI ArcGIS, Marine AIS	`OGC WFS 2.0 GeoJSON`, `PostGIS`, AIS NMEA/JSON
5G RAN, GPON & IP Core	Ericsson ENM, Cisco IOS-XR, Juniper, GPON OLTs	`3GPP TS 32.435 XML`, `ITU-T G.984`, Syslog, SNMPv2c
ITSM, DCIM & BSS	ServiceNow FSM/ITSM, Jira SM, NetBox, Comviva	`ServiceNow Table API`, `Jira REST v3`, `TMF629 / TMF638`

Sovereign & Cloud Deployment: Runs identically via `Docker Compose` or `Kubernetes (OpenShift / EKS / AKS)`. Supports zero-PII air-gapped telco data centres using local `vLLM + bge-m3 + ONNX` inference, and ships with a standalone Multi-OEM Lab Emulator (`:8001`) for offline stress testing.